

Unsupervised Learning: A Comprehensive Guide to Clustering

Introduction

Unsupervised learning is a type of machine learning where the model is trained on data without labelled responses. The goal is to uncover hidden patterns, structures, or relationships within the data. One of the most common techniques in unsupervised learning is **clustering**, which groups similar data points together based on their features.

This document provides an in-depth overview of clustering, its key concepts, algorithms, evaluation metrics, and applications. It is designed to be beginner-friendly while maintaining a high standard of quality, ensuring that new learners can understand and apply these concepts effectively.

Key Definitions

1. Unsupervised Learning

- **Definition:** A machine learning paradigm where the algorithm learns patterns from unlabelled data. Unlike supervised learning, there are no target labels or outputs provided.
- **Goal:** To discover the underlying structure of the data, such as grouping similar data points or reducing the dimensionality of the data.

2. Clustering

- **Definition:** A technique used to group a set of objects in such a way that objects in the same group (called a **cluster**) are more similar to each other than to those in other groups.
- **Applications:** Customer segmentation, image compression, anomaly detection, and more.

3. Feature

- **Definition:** An individual measurable property or characteristic of a phenomenon being observed. For example, in a dataset of tumours, features could include size, shape, and texture.

4. Distance Metric

- **Definition:** A function that defines the distance between two data points. Common distance metrics include:
 - **Euclidean Distance:** The straight-line distance between two points in Euclidean space.
 - **Manhattan Distance:** The sum of absolute differences between coordinates.
 - **Cosine Similarity:** Measures the cosine of the angle between two vectors.

5. Centroid

- **Definition:** The centre of a cluster, often calculated as the mean of all points in the cluster.

6. Dimensionality Reduction

- **Definition:** The process of reducing the number of random variables (features) under consideration by obtaining a set of principal variables. Common techniques include:
 - **Principal Component Analysis (PCA):** A linear dimensionality reduction technique that transforms data into a lower-dimensional space while preserving as much variance as possible.
 - **t-SNE (t-Distributed Stochastic Neighbour Embedding):** A non-linear dimensionality reduction technique used for visualizing high-dimensional data in 2D or 3D.

7. Outlier

- **Definition:** A data point that significantly differs from other data points in the dataset. Outliers can be caused by variability in the data or errors.

8. Noise

- **Definition:** Irrelevant or meaningless data points that do not contribute to the underlying pattern or structure.

9. Dendrogram

- **Definition:** A tree-like diagram used to visualize the arrangement of clusters produced by hierarchical clustering.

10. Soft Clustering

- **Definition:** A clustering approach where each data point can belong to multiple clusters with varying degrees of membership (e.g., Gaussian Mixture Models).

11. Hard Clustering

- **Definition:** A clustering approach where each data point belongs to exactly one cluster (e.g., K-Means).

Clustering Algorithms

1. K-Means Clustering

- **Definition:** A partitioning method that divides the dataset into **k** clusters, where each data point belongs to the cluster with the nearest centroid.
- **Steps:**
 1. Choose the number of clusters (**k**).
 2. Initialize centroids randomly.
 3. Assign each data point to the nearest centroid.
 4. Recalculate the centroids based on the assigned points.
 5. Repeat until convergence (when centroids no longer change significantly).
- **Advantages:** Simple and efficient for large datasets.
- **Limitations:** Requires the number of clusters (**k**) to be specified in advance; sensitive to initial centroid placement.

2. Hierarchical Clustering

- **Definition:** A method that builds a hierarchy of clusters, either by merging smaller clusters into larger ones (**agglomerative**) or by splitting larger clusters into smaller ones (**divisive**).
- **Steps:**
 1. Start with each data point as its own cluster.
 2. Merge the closest pairs of clusters iteratively.
 3. Repeat until all data points are in a single cluster.
- **Advantages:** Does not require the number of clusters to be specified in advance; produces a dendrogram for visualization.
- **Limitations:** Computationally expensive for large datasets.

3. DBSCAN (Density-Based Spatial Clustering of Applications with Noise)

- **Definition:** A density-based clustering algorithm that groups together points that are closely packed, marking points in low-density regions as outliers.
- **Key Parameters:**
 - **Epsilon (ϵ):** The maximum distance between two points to be considered neighbours.
 - **MinPts:** The minimum number of points required to form a dense region.
- **Advantages:** Can find arbitrarily shaped clusters and identify outliers.
- **Limitations:** Struggles with clusters of varying densities.

4. Gaussian Mixture Models (GMM)

- **Definition:** A probabilistic model that assumes all data points are generated from a mixture of a finite number of Gaussian distributions with unknown parameters.
- **Advantages:** Provides soft clustering (probabilistic assignments to clusters).
- **Limitations:** Computationally intensive and sensitive to initialization.

5. Mean Shift Clustering

- **Definition:** A centroid-based algorithm that iteratively shifts centroids toward the highest density of data points.
 - **Advantages:** Does not require the number of clusters to be specified.
 - **Limitations:** Computationally expensive for large datasets.
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Evaluation Metrics for Clustering

1. Silhouette Score

- **Definition:** A metric that measures how similar an object is to its own cluster compared to other clusters. Scores range from -1 to 1, where higher values indicate better-defined clusters.
- The score is higher when clusters are dense and well separated, which relates to a standard concept of a cluster.
- The Silhouette Coefficient is generally higher for convex clusters than other concepts of clusters, such as density-based clusters like those obtained through DBSCAN.

2. Inertia

- **Definition:** The sum of squared distances of data points to their closest centroid. Lower inertia indicates better clustering.

3. Davies-Bouldin Index

- **Definition:** A metric that evaluates the average similarity ratio of clusters. Lower values indicate better clustering.
- Zero is the lowest possible score. Values closer to zero indicate a better partition.
- The computation of Davies-Bouldin is simpler than that of Silhouette scores.
- The index is solely based on quantities and features inherent to the dataset as its computation only uses point-wise distances.
- The Davies-Bouldin index is generally higher for convex clusters than other concepts of clusters, such as density-based clusters like those obtained from DBSCAN.
- The usage of centroid distance limits the distance metric to Euclidean space.

4. Calinski-Harabasz Index

- **Definition:** A metric that measures the ratio of between-cluster dispersion to within-cluster dispersion (*where dispersion is defined as the sum of distances squared*).
- Also known as the Variance Ratio Criterion- Higher values indicate better clustering.
- The score is higher when clusters are dense and well separated, which relates to a standard concept of a cluster.
- The score is fast to compute.
- The Calinski-Harabasz index is generally higher for convex clusters than other concepts of clusters, such as density-based clusters like those obtained through DBSCAN.

5. Adjusted Rand Index (ARI)

- **Definition:** A metric that measures the similarity between two clustering, adjusted for chance. Scores range from -1 to 1, where higher values indicate better agreement.
- As with all clustering metrics, one can permute 0 and 1 in the predicted labels, rename 2 to 3, and get the same score.
- Perfect labelling is scored 1.0
- Poorly agreeing labels have lower scores, and for the adjusted Rand index the score will be negative or close to zero.
- However, for the unadjusted Rand index the score, while lower, will not necessarily be close to zero

6. Mutual Information (MI)

- **Definition:** A metric that measures the mutual dependence between two clustering. Higher values indicate better agreement.
- Two different normalized versions of this measure are available, Normalized Mutual Information (NMI) and Adjusted Mutual Information (AMI). NMI is often used in the literature, while AMI was proposed more recently and is normalized against chance.
- One can permute 0 and 1 in the predicted labels, rename 2 to 3 and get the same score.
- All, **mutual_info_score**, **adjusted_mutual_info_score** and **normalized_mutual_info_score** are symmetric: swapping the argument does not change the score.
- Perfect labelling is scored 1.0; this is not true for **mutual_info_score**, which is therefore harder to judge.

7. Homogeneity, Completeness, and V-measure

- **Homogeneity:**
 - Each cluster contains only members of a single class.
 - Measures how "pure" the clusters are.
- **Completeness:**
 - All members of a given class are assigned to the same cluster.
 - Measures how well all data points of a class are grouped together.
- **V-measure:**
 - The harmonic means of homogeneity and completeness.
 - **Formula:**
$$V\beta = \frac{(1+\beta) \cdot \text{homogeneity} \cdot \text{completeness}}{(\beta \cdot \text{homogeneity} + \text{completeness})}$$
 - **Range:** 0.0 (worst) to 1.0 (perfect).
- **Beta Parameter:**
 - Controls the weight between homogeneity and completeness.
 - **Default:** $\beta=1$ (equal weight).
 - $\beta<1$: More weight to homogeneity.
 - $\beta>1$: More weight to completeness.

8. Fowlkes-Mallows scores

The Fowlkes-Mallows score FMI is defined as the geometric mean of the pairwise precision and recall.

- The score ranges from 0 to 1. A high value indicates a good similarity between two clusters.
- One can permute 0 and 1 in the predicted labels, rename 2 to 3 and get the same score.
- Perfect labelling is scored 1.0
- Bad (e.g. independent labelling) have zero scores.

Applications of Clustering

1. **Customer Segmentation:** Grouping customers based on purchasing behaviour.
2. **Image Segmentation:** Dividing an image into regions with similar properties.
3. **Anomaly Detection:** Identifying unusual patterns in data.
4. **Document Clustering:** Grouping similar documents for topic modelling.
5. **Gene Expression Analysis:** Grouping genes with similar expression patterns.

Conclusion

Clustering is a powerful unsupervised learning technique that helps uncover hidden structures in data. By understanding the key concepts, algorithms, and evaluation metrics, you can apply clustering to solve real-world problems effectively. Experiment with different algorithms and evaluation metrics to find the best approach for your dataset.